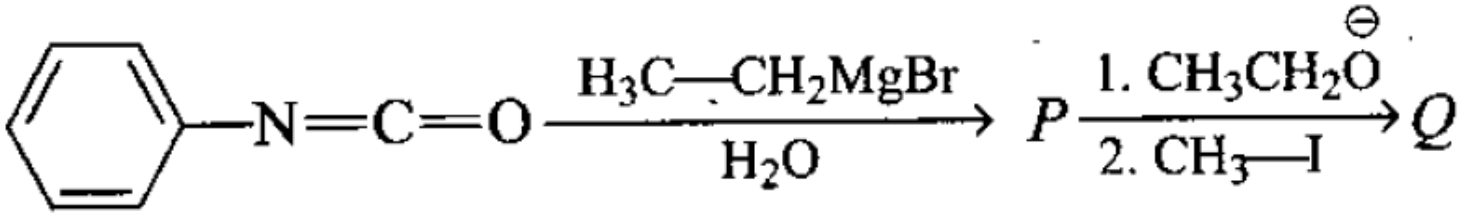
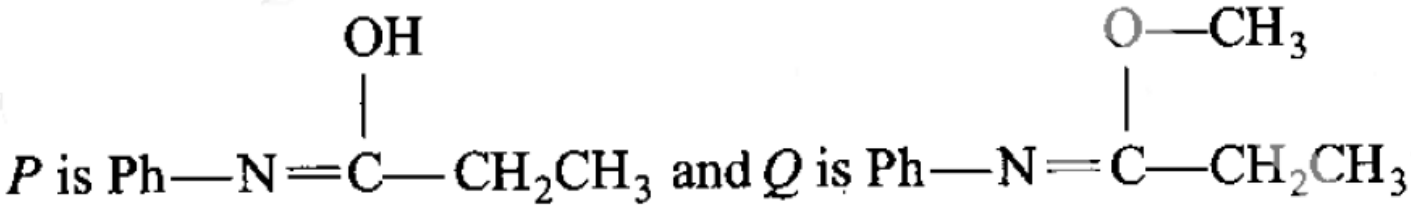
Q.1

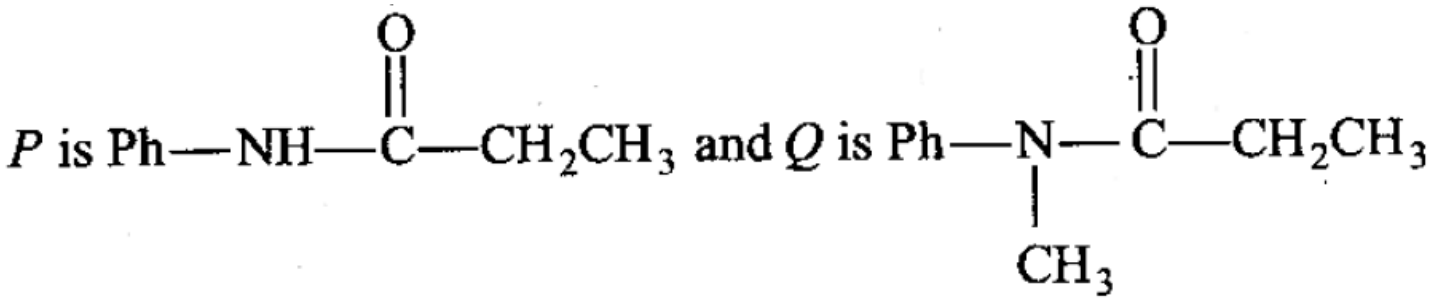


Find out P and Q :

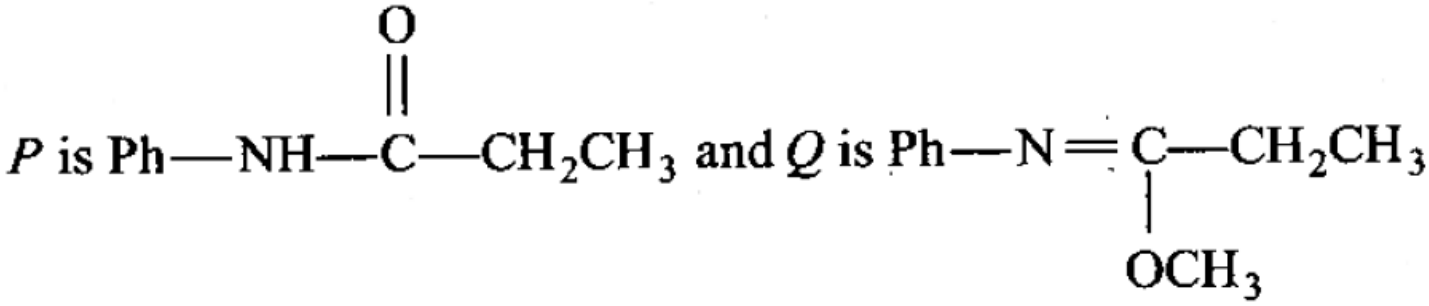
(A)



(B)



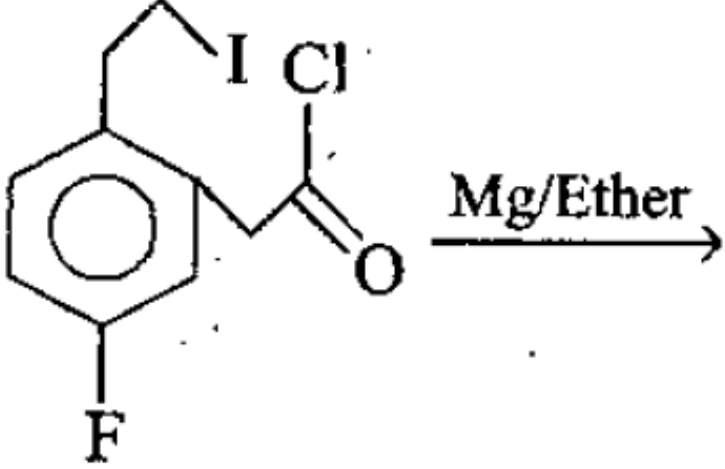
(C)



(D) P and Q both are same

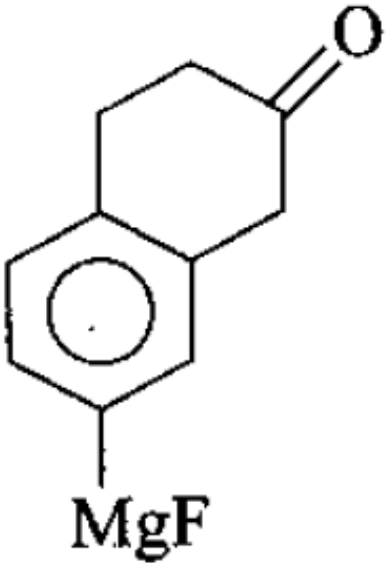
Q.2

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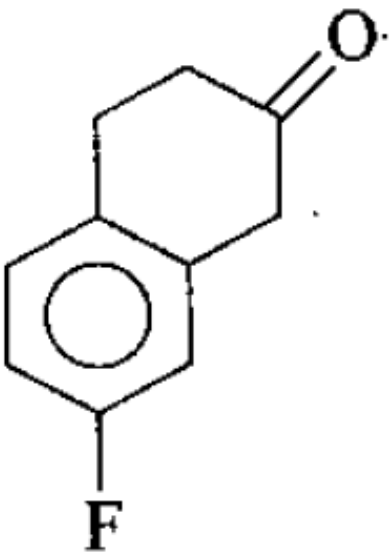


The final product of the reaction is :

(A)

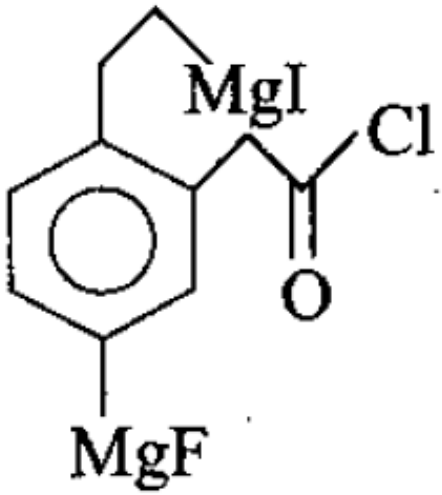


(B)

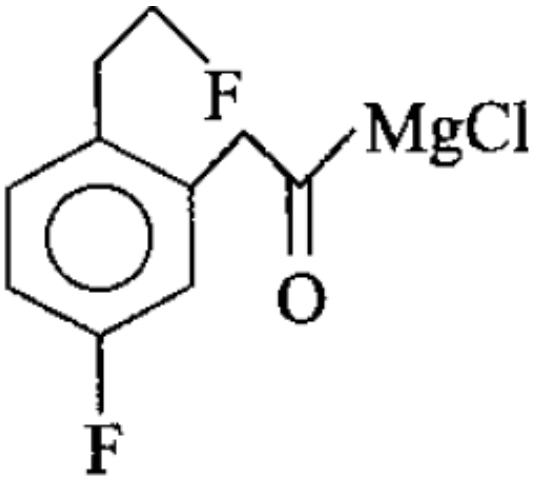


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(C)



(D)

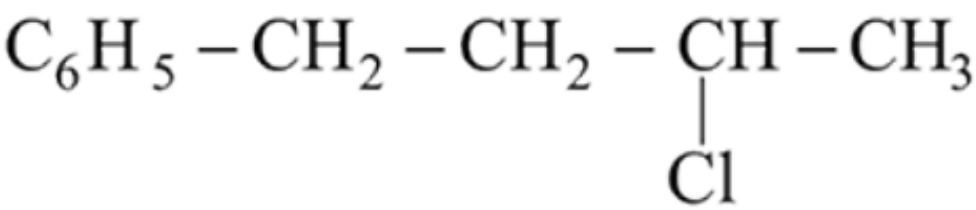


Q.3

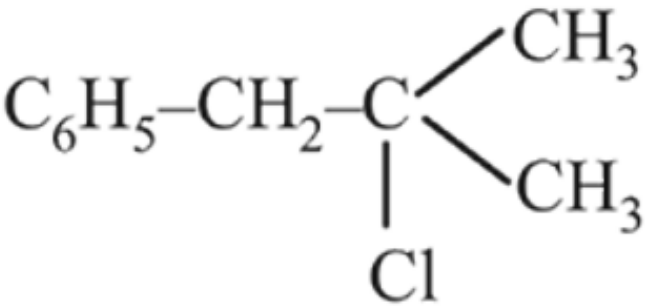
A compound A with molecular formula $C_{10}H_{13}Cl$ gives a white precipitate on adding silver nitrate solution. A on reacting with alcoholic KOH gives compound B as the main product. B on ozonolysis gives C and D. C gives Cannizaro reaction but not aldol condensation. D gives aldol condensation but not Cannizaro reaction. A is:

(A) $C_6H_5 - CH_2 - CH_2 - CH_2 - CH_2 - Cl$

(B)

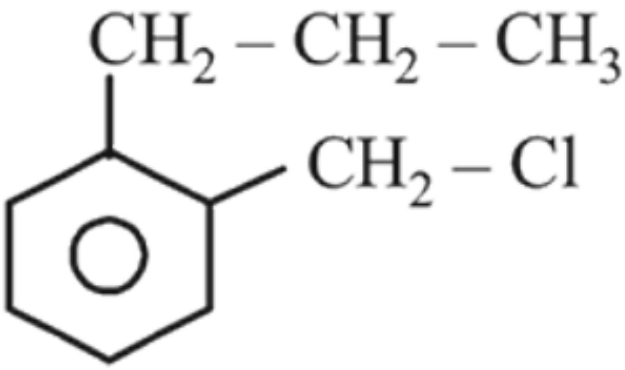


(C)



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(D)



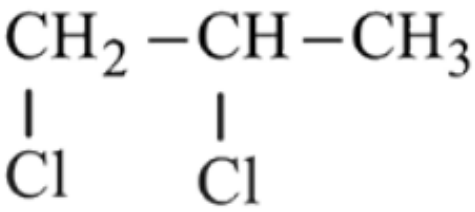
Q.4

In the following reaction sequence :

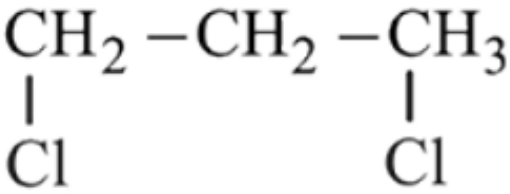
$$\underset{(C_3H_6Cl_2)}{I} \xrightarrow{KOH(aq)} II \xrightarrow[(ii)\,H_2O/H^+]{(i)\,CH_3MgBr} III \xrightarrow{Anhy.\,ZnCl_2 + Conc.\,HCl} \text{given turbidity immediately}$$

The compound I is :

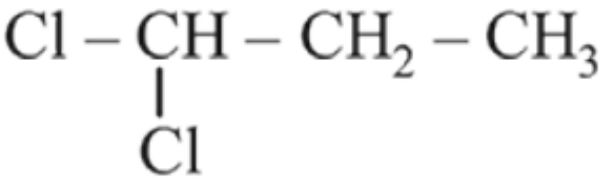
(A)



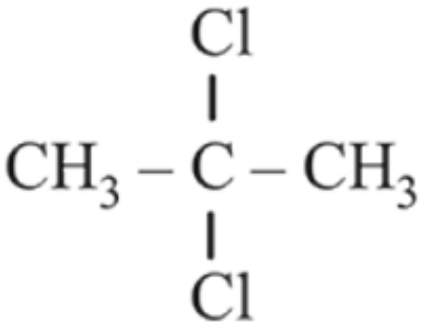
(B)



(C)



(D)



Q.5

Which of the following will have a meso isomer?

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(A) 2,3- Dichloropentane

(B) 2,3 -Dichlorobutane

(C) 2-Chlorobutane

(D) 2-Hydroxypropanoic acid

Q.6

Conversion of benzene diazonium chloride to chlorobenzene is an example of which of the following reactions?

(A) Claisen

(B) Friedel-craft

(C) Sandmeyer

(D) Wurtz

Q.7

Arrange the following compounds in order of increasing dipole moment :
(I) Toluene
(II) m -dichlorobenzene
(III) o -dichlorobenzene
(IV) p -dichlorobenzene

(A) $\mathrm{I} < \mathrm{IV} < \mathrm{II} < \mathrm{III}$

(B) $\mathrm{IV} < \mathrm{I} < \mathrm{II} < \mathrm{III}$

(C) $\mathrm{IV} < \mathrm{I} < \mathrm{III} < \mathrm{II}$

(D) $\mathrm{IV} < \mathrm{II} < \mathrm{I} < \mathrm{III}$

Q.8

The Wurtz-Fittig reaction involves condensation of

(A) two molecules of aryl halides.

(B) one molecule of each of aryl-halide and alkyl-halide.

(C) one molecule of each of aryl-halide and phenol.

(D) two molecules of aralkyl-halides.

Q.9

Aryl fluoride may be prepared from arene diazonium chloride using :

(A) HBF_4/Δ

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(B) $HBF_4/NaNO_2, Cu, \Delta$

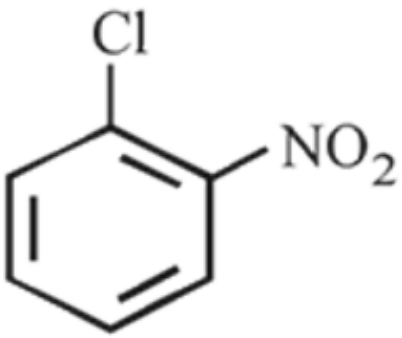
(C) CuF/HF

(D) Cu/HF

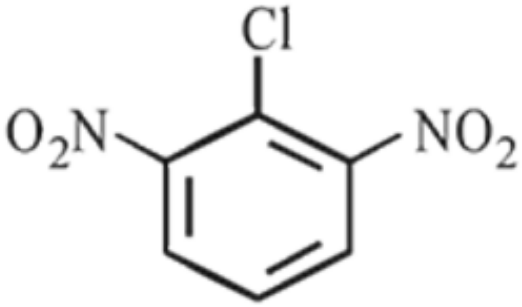
Q.10

A major component of Borsch reagent is obtained by reacting hydrazine hydrate with which of the following?

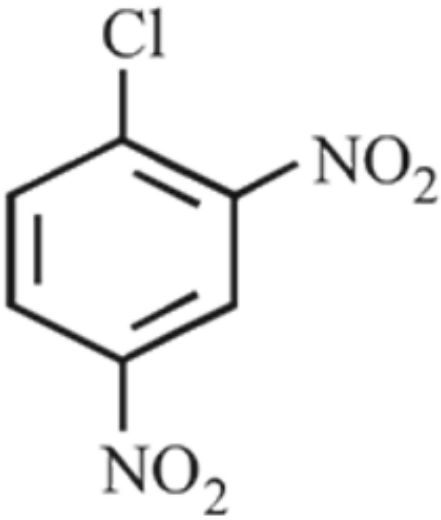
(A)



(B)

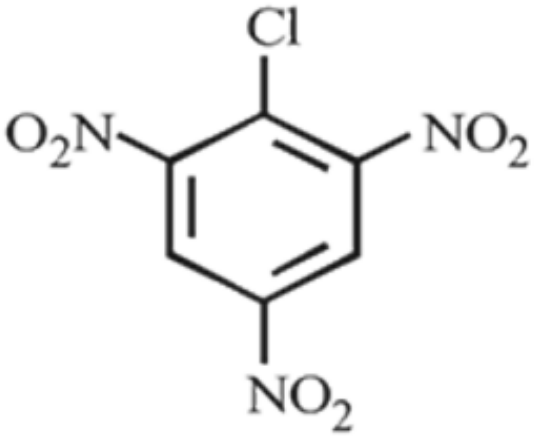


(C)



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(D)



Q.11

Replacement of Cl of chlorobenzene to give phenol requires drastic conditions but chlorine of 2,4 - dinitrochlorobenzene is readily replaced This is because

(A) NO_2 makes the ring electron rich at ortho and para positions.

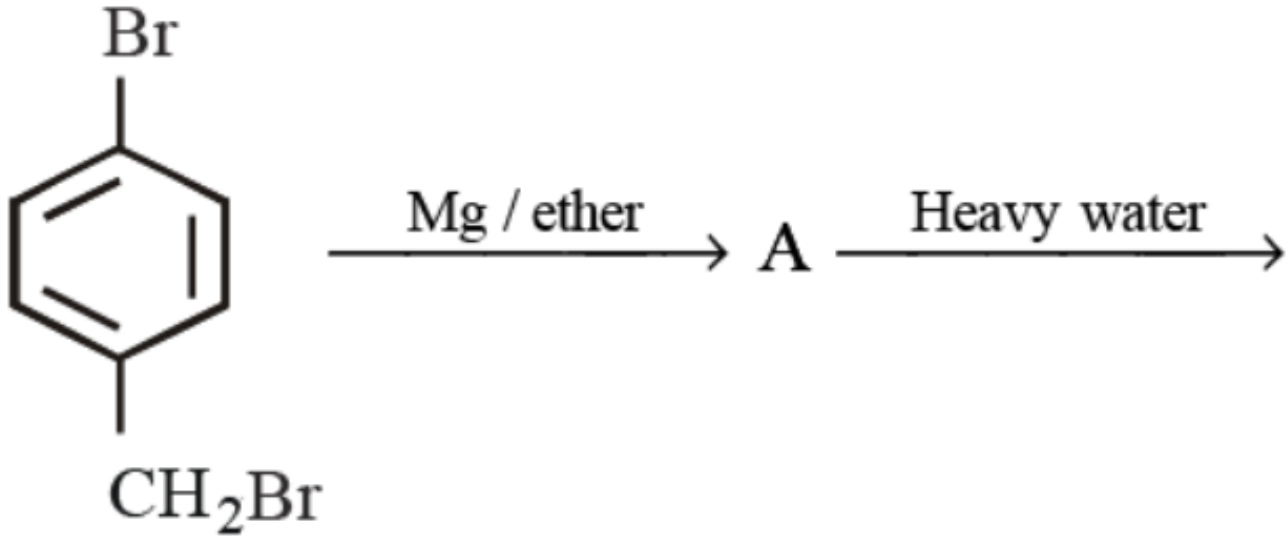
(B) NO_2 withdraws e^- from meta-position.

(C) NO_2 donates e^- at meta-position.

(D) NO_2 withdraws e^- from ortho/para-positions.

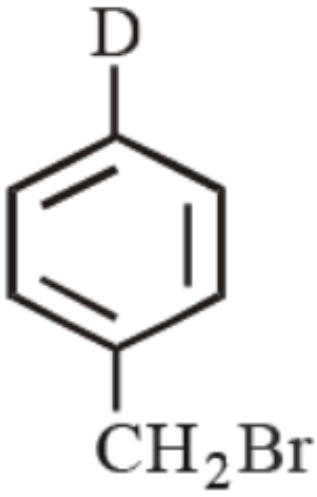
Q.12

The final product obtained in the reaction

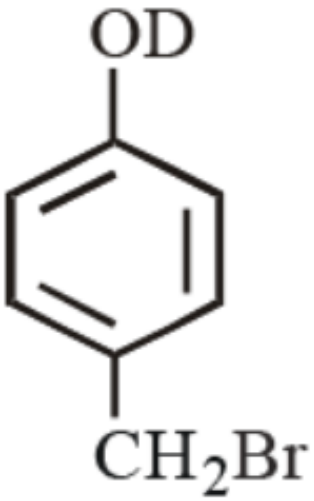


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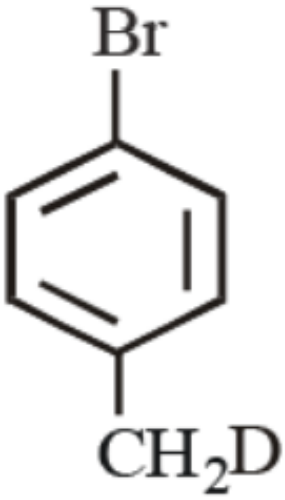
(A)



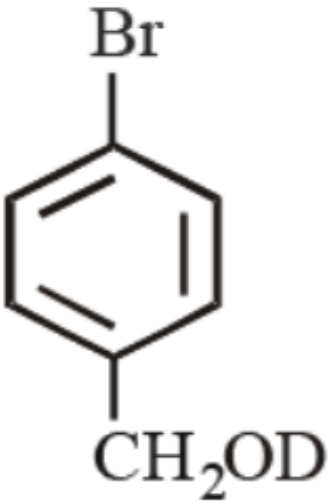
(B)



(C)



(D)



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Q.13

The compound

$$C_7H_8 \xrightarrow{3Cl_2/\Delta} A \xrightarrow{Br_2/Fe} B \xrightarrow{Zn/HCl} C$$

The compound C is

(A) o -Bromotoluene

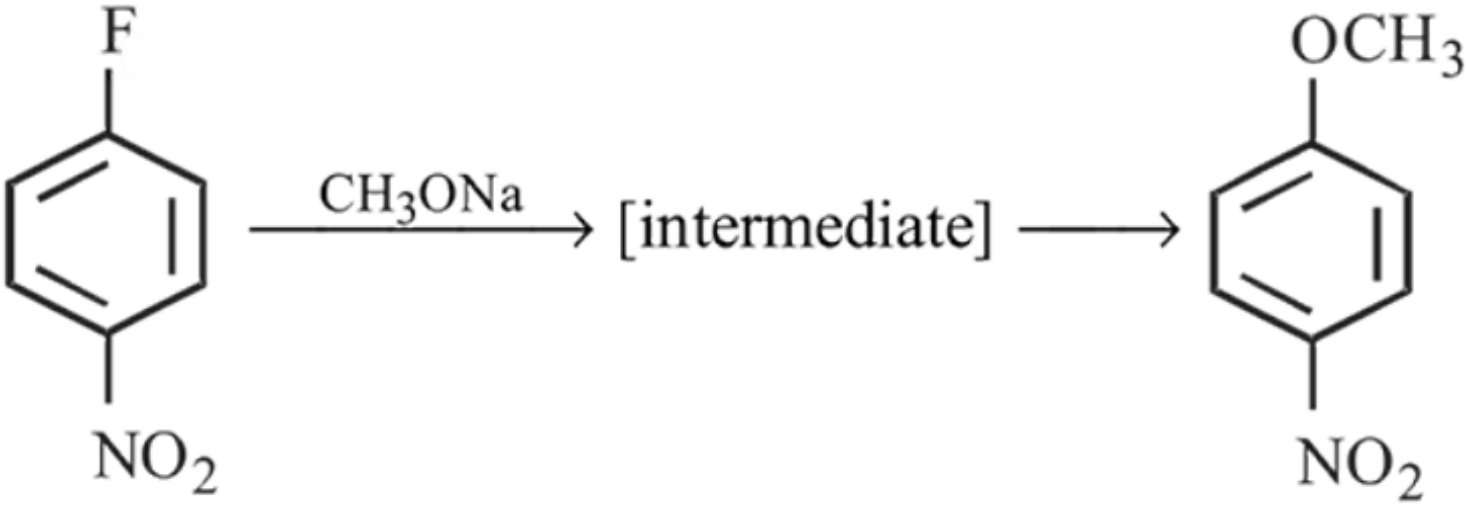
(B) m -Bromotoluene

(C) p -Bromotoluene

(D) 3-Bromo-2, 4,6 -trichlorotoluene

Q.14

Which of the following is (are) true concerning the intermediate in the addition-elimination mechanism of the following reaction ?



A = The intermediate is aromatic, B = The intermediate is a resonance stabilised anion, C = Electron withdrawing groups on the benzene
ring stabilises the intermediate

(A) only A

(B) only B

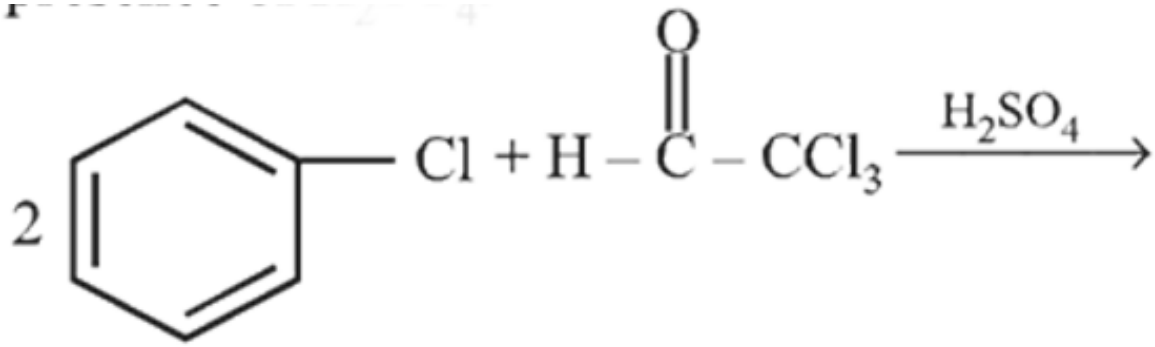
(C) A and C

(D) B and C

Q.15

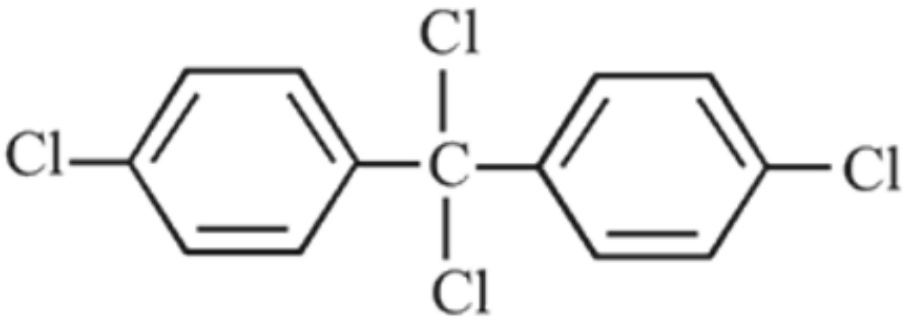
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Chlorobenzene reacts with trichloro acetaldehyde in the presence of H_2SO_4.

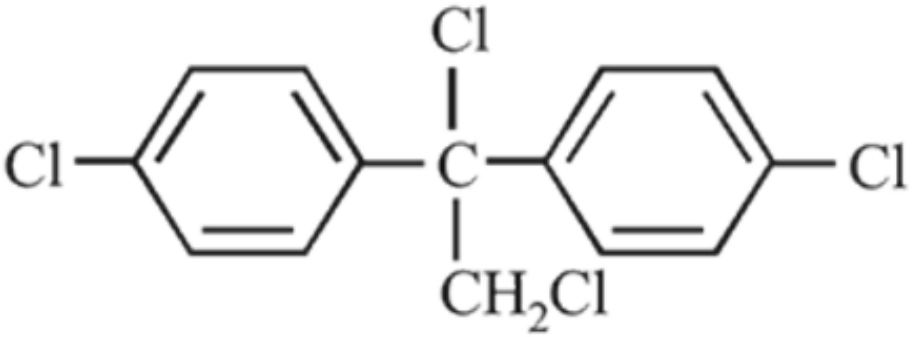


The major product formed is:

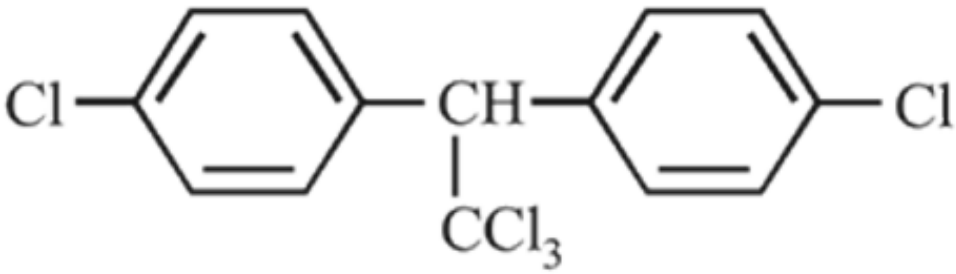
(A)



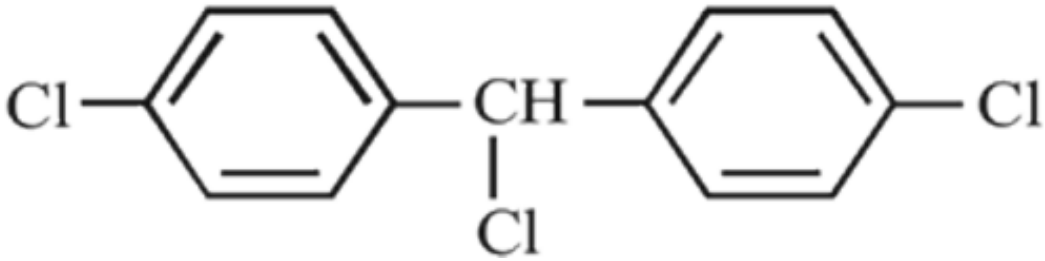
(B)



(C)



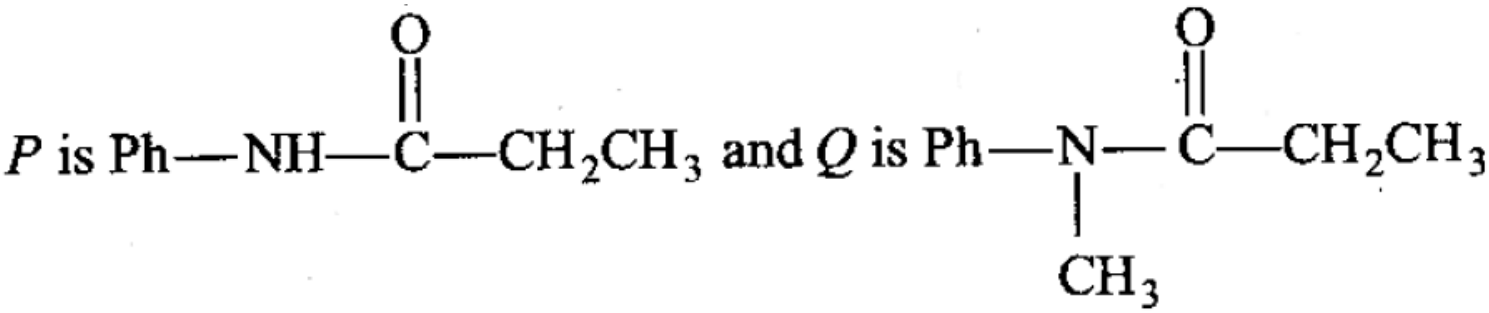
(D)



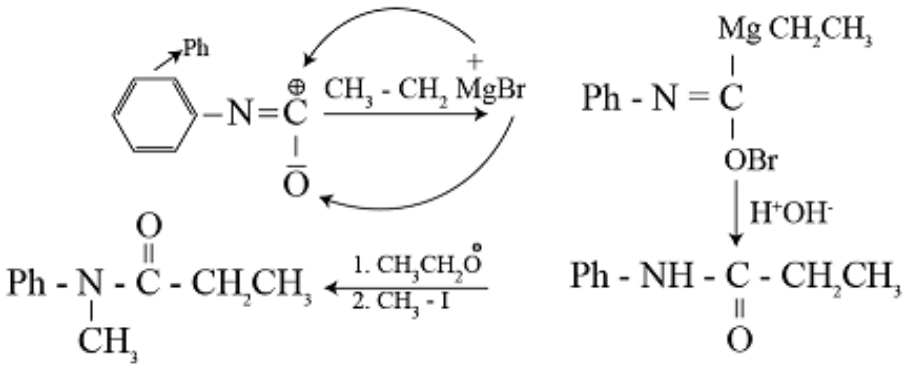
Answers & Solutions

Q.1 Answer:

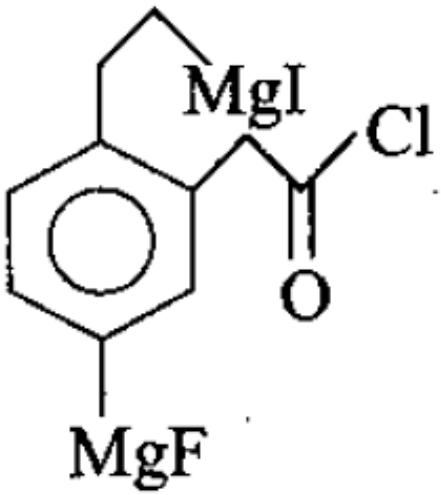
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Solution:



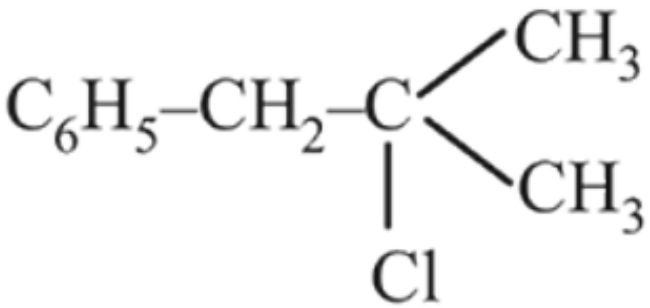
Q.2 Answer:



Solution:

No solution available

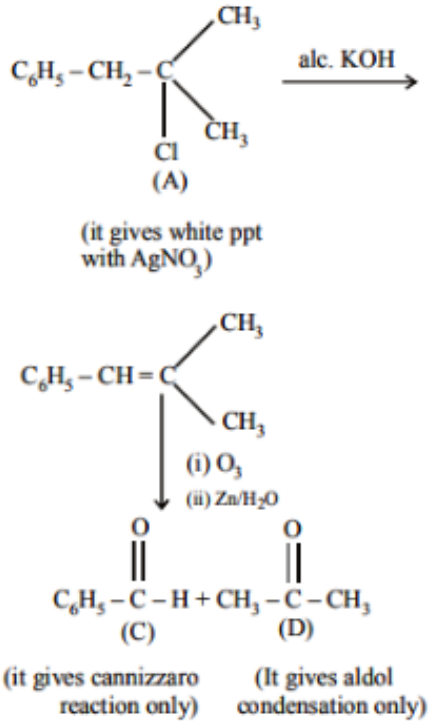
Q.3 Answer:



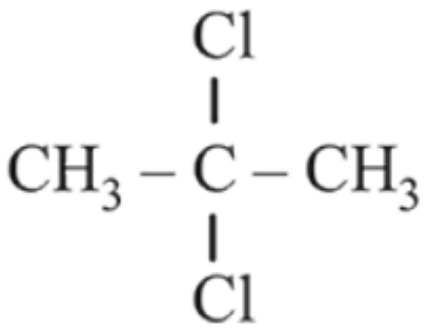
Solution:

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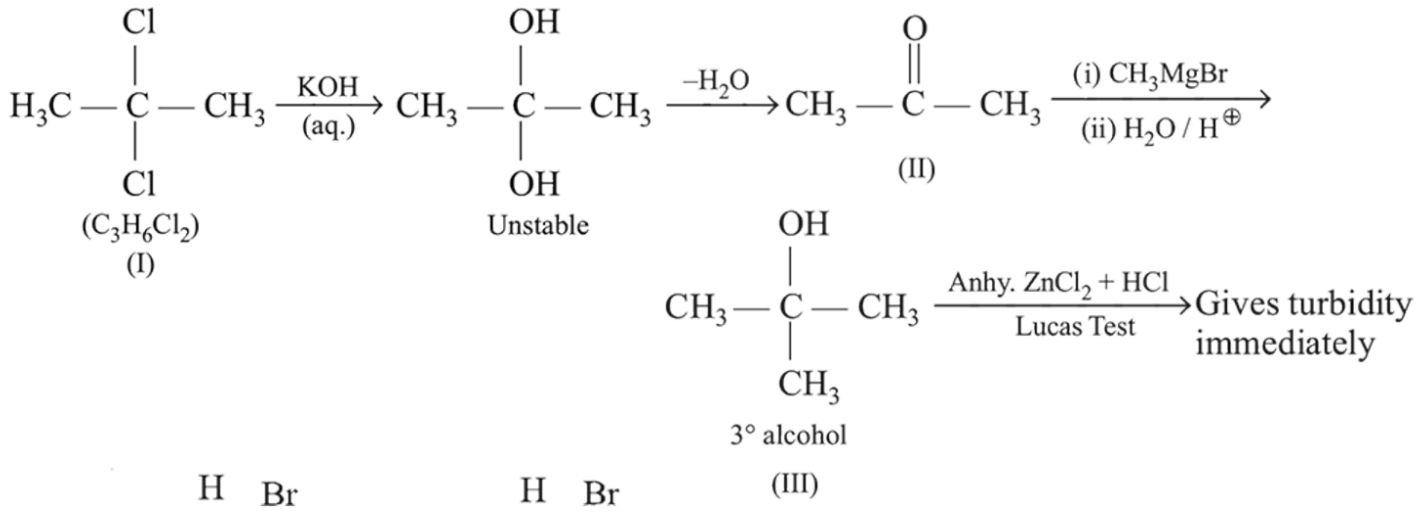
Compound A reacts with alc.KOH to give compound B which on further ozonolysis gives C (does not contains a- H atom) and D (contains a $-$ H atom).This reaction sequence can be achieved by compounds in option (a) and (c). Since compound A gives white $ppt.$ with $AgNO_3$ preferable option will be (c) as tert alkyl reacts with $AgNO_3$ more quickly



Q.4 Answer:



Solution:



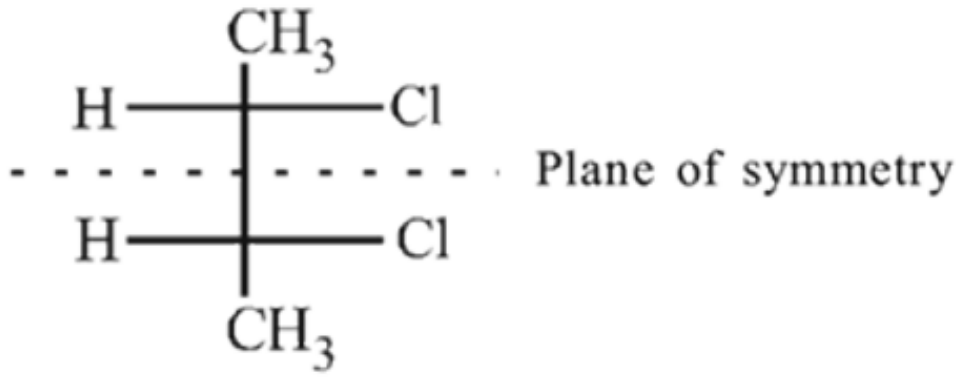
Q.5 Answer:

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2,3 -Dichlorobutane

Solution:

The compound has two similar assymmetric C-atoms. It has plane of symmetry and exist in meso form.

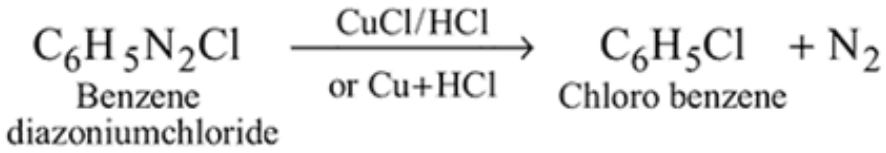


Meso - 2, 3 dichlorobutane

Q.6 Answer:

Sandmeyer

Solution:



The above reaction is known as Sandmayer's reaction.

Q.7 Answer:

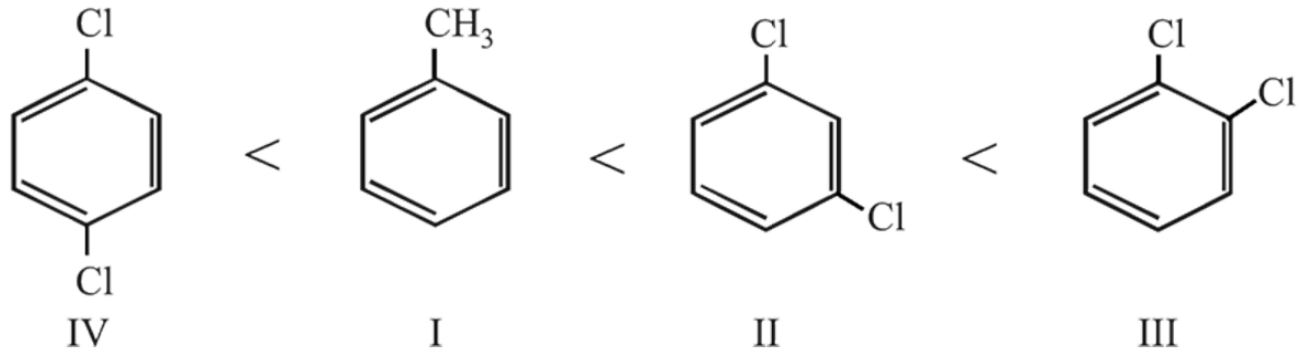
$\mathrm{IV} < \mathrm{I} < \mathrm{II} < \mathrm{III}$

Solution:

In p -dichlorobenzene, the two equal dipoles are in opposite direction, hence the molecule has zero dipole moment. In o - and m - dichlorobenzenes, the two dipoles are at 60° and 120° apart respectively, and thus according to parallelogram law of forces, the dipole moment of o - dichlorobenzene is much higher than that of m -isomer. Lastly, toluene with a +I group possesses

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little dipole moment. Thus the overall order is



Q.8 Answer:

one molecule of each of aryl-halide and alkyl-halide.

Solution:

Reaction between alkyl halide, aryl halide and sodium in presence of ether is known as Wurtz fitting reaction

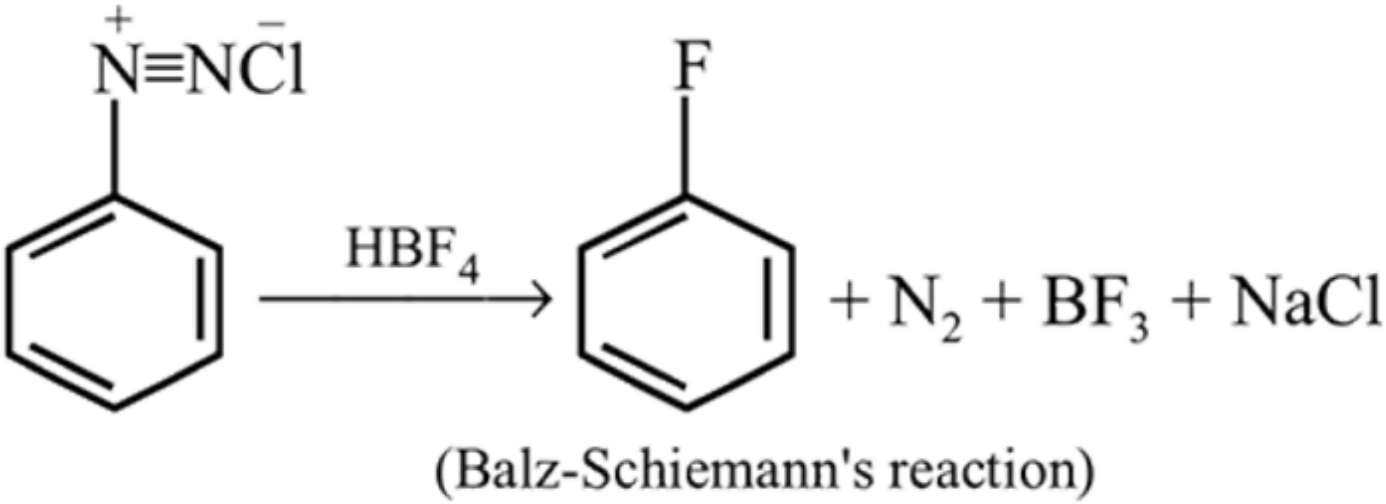
$$C_6H_5Cl + 2Na + ClCH_3 \longrightarrow C_6H_5CH_3 + 2NaCl$$

Toluene

Q.9 Answer:

HBF_4/Δ

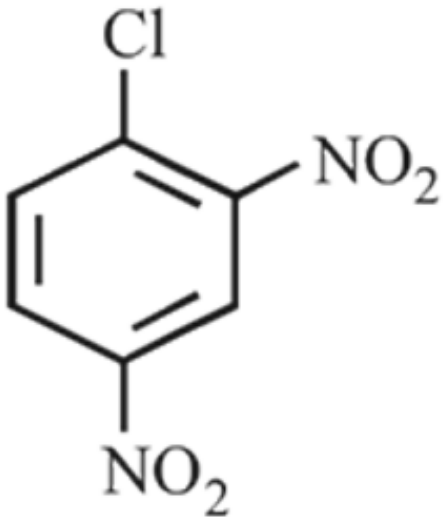
Solution:



(Balz-Schiemann's reaction)

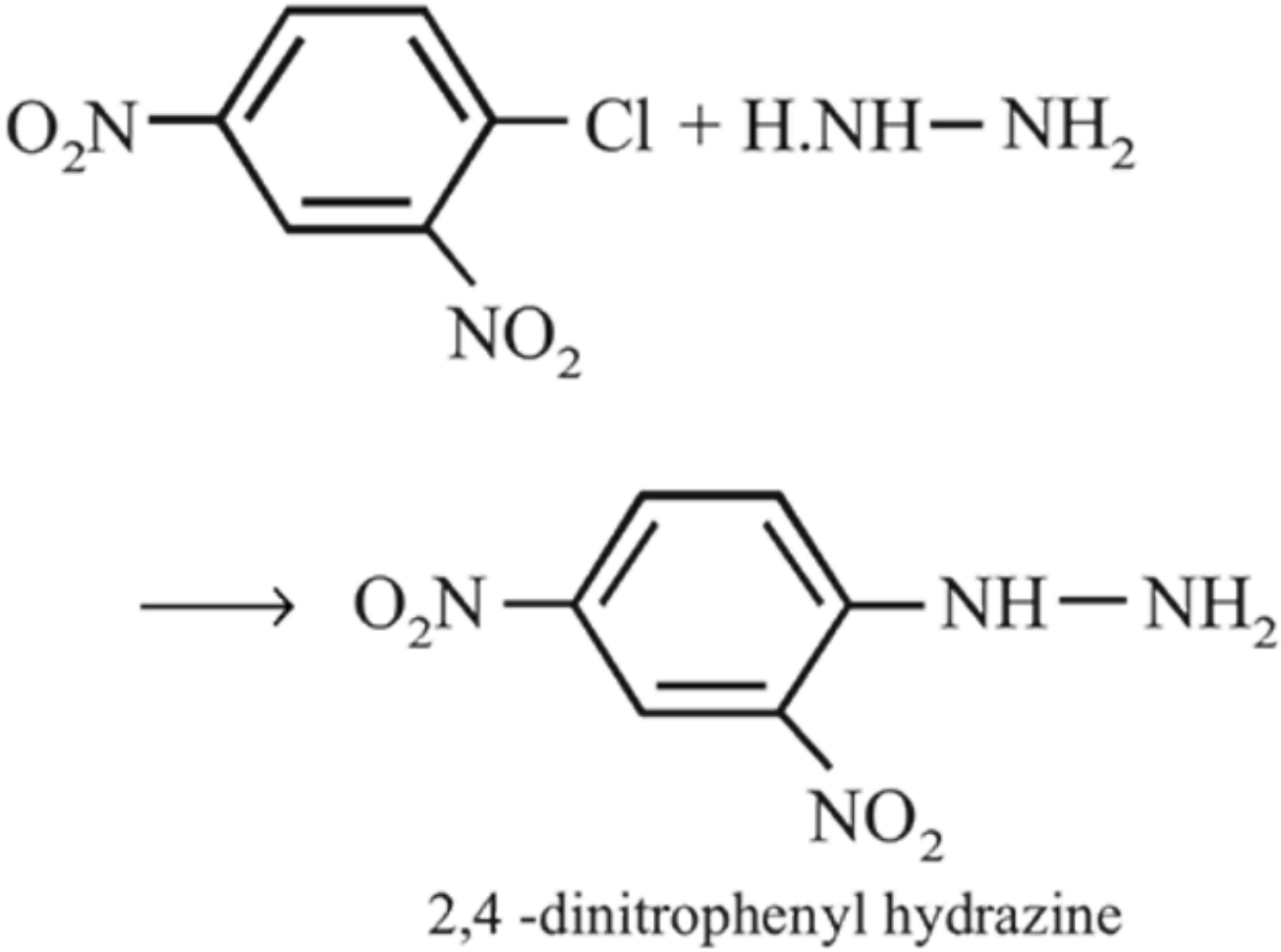
Q.10 Answer:

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Solution:

The major component of Borsch reagent is 2,4- dinitrophenyl hydrazine which can be obtained by reaction of 2,4 -dinitrochloro benzene and hydrazine



2,4 -dinitrophenyl hydrazine

Q.11 Answer:

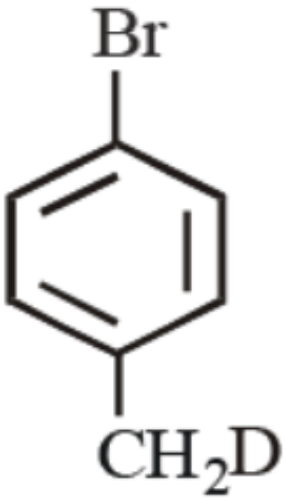
NO_2 withdraws e^- from ortho/para-positions.

Solution:

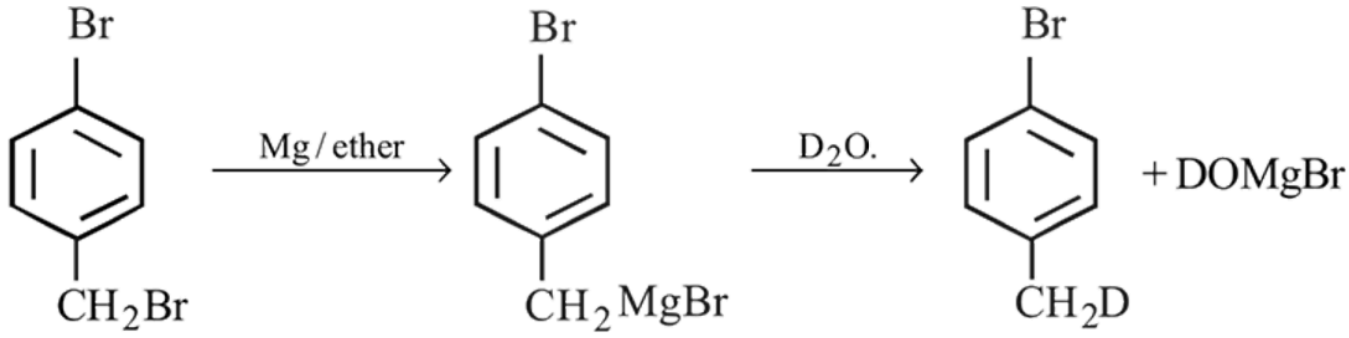
$-NO_2$ group withdraws electrons from *o* - and *p* -positions and hence activates the $-Cl$ present on that position towards nucleophilic substitution.

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Q.12 Answer:



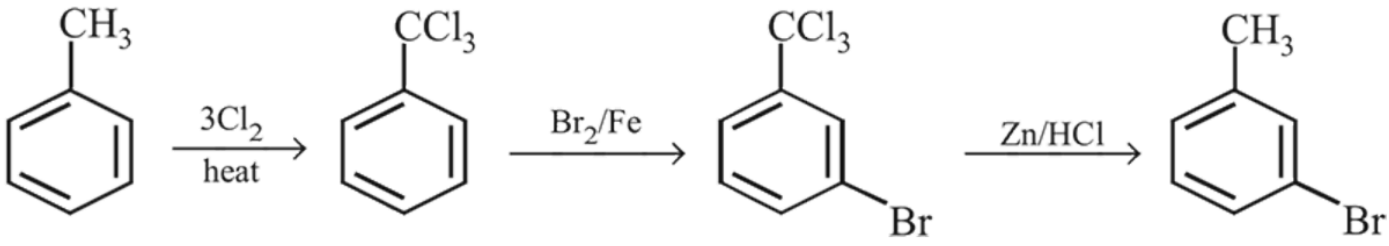
Solution:



Q.13 Answer:

m -Bromotoluene

Solution:

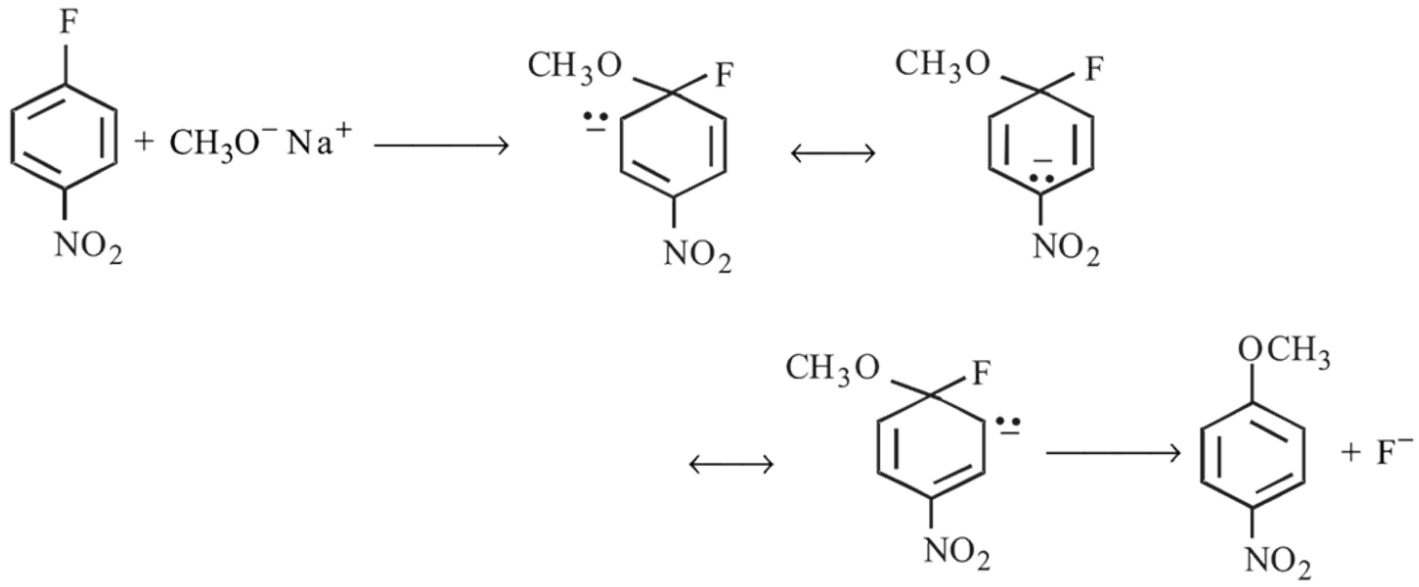


Q.14 Answer:

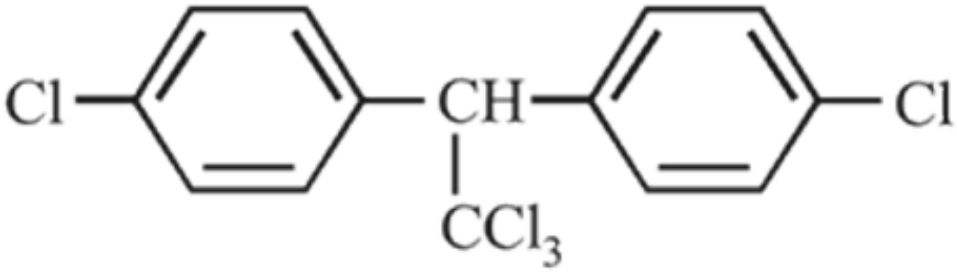
B and C

Solution:

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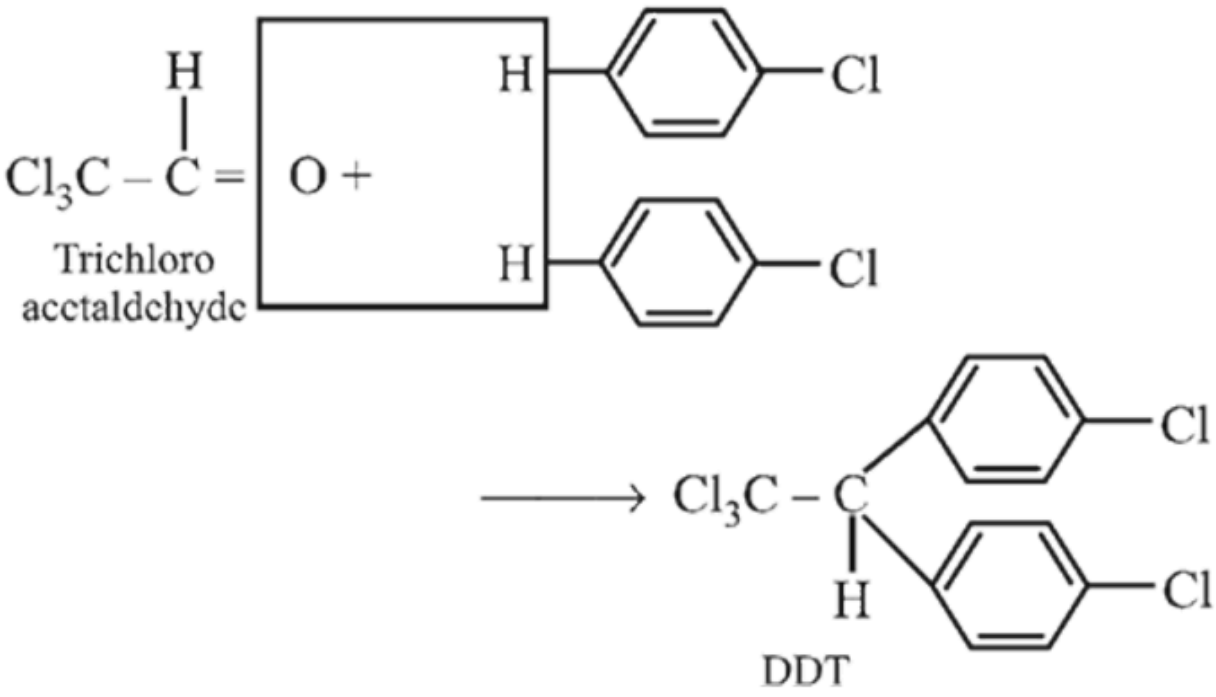


Q.15 Answer:



Solution:

Chloral on reaction with chlorobenzene in the presence of a catalytic amount of sulphuric acid forms DDT (dichlorodiphenyl Trichloro ethane).



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